

Critical Analysis of Edge Computing

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Abstract—The edge computing is facing many problems and efforts are being made to overcome those challenges. Hybrid Mobile Edge Computing is introduced for the mobile devices to overcome the limited battery issues and performance constraints in the devices. There were some difficulties in the integrated development environment of the edge computing, and to overcome those problems, a container-based method is introduced which improves the performance of the coding environment in EC, and also it facilitates in-place debugging. An EC architecture is presented that provides local data processing, management, and quick reaction for the Virtual IoT Devices. With EC's support, a hybrid computing framework is built and an intelligent resource scheduling strategy to fulfill the real-time requirements in smart manufacturing; which showed satisfactory results. A Multi-source Transmission Protocol is presented to counter problems such as the low video streaming and high bandwidth usage. An ECD device is presented for the displaying of results in real-time, with short-time response and also to overcome the network limits. A decentralized resource management technique is introduced along with a technical framework for the latency-sensitive applications' deployment on the edge devices while protecting the privacy of devices. A Dynamic Edge-Fabric Environment is presented; this platform can automatically learn on the basis of past performance of the available resources using machine learning techniques and it decides which task should be best executed where, based on real-time system status and task requirements; the results have proven that it can improve overall performance for the selection of resources.

Keywords—Edge Computing, Internet of Things, Mobile Applications, Remote Debugging, Container, Docker, Local Data Processing, Virtual IoT Device, Smart Manufacturing, Resource Scheduling, Multi-source download, DCU, Decentralization, Resource Management

I. INTRODUCTION

We are living in the 21st century, an era in which almost every successful business is using high-end technologies to improve and perform their business operations in rapid acceleration. If we go about 8 years back in time, we find

many businesses and individuals embracing the cloud technologies and the Internet of Things (IoT) [1]. The cloud technologies [2], where it is very helpful, has also proven to be otherwise—or to be accurate, not as much efficient, when it comes to using it in the context of the Internet of Things. What happened was that, earlier, the businesses and/or the individuals started using IoT enabled devices to improve their operations. And the IoT devices required storage as the base to store their data for processing on it. This data was kept on a server, which we called cloud storage. The cloud storage has its benefits, one can read it from the provided reference at the end of this paper. So, this processing, i.e. fetching data from the cloud technologies to perform computations on it (i.e. Cloud Computing), created some problems such as low latency, and security, limited scalability, and unreliability, etc. To tackle all these issues, edge computing was introduced which not only turned these issues into benefits, add to that, it also paved the way for many other business opportunities [3], [4]. By bringing the data centers closer to the devices (i.e. bringing the data centers local), companies can now easily target their desired markets, giving them the advantage of versatility, without having to spend on the expansion of their infrastructure. Edge computing is taking technology to the next level. According to Forbes it is forecasted that the global edge computing market will reach 1.12 trillion dollars by the year 2023.¹

Below are the different novel ideas that demonstrate the current progress relating to the edge computing.

- 1) Edge computing helped in reducing the latency problem, but it created some new issues like security (because there is no central device anymore), and performance and battery issues in the mobile devices. To eliminate these issues, Hybrid Mobile Edge Computing (HMEC) model is introduced. In the future, owners will have full control over the network and the data will only be sent to the secure nodes, and be prevented from the insecure nodes, by the use of integration policy. And thus the extended

¹ https://www-forbes-com.cdn.ampproject.org/v/s/www.forbes.com/sites/cognitiveworld/2020/04/16/edge-ai-is-the-future-intel-and-udacity-are-teaming-up-to-train-developers/amp?amp_js_v=a3&_gsa=l&usqp=mq331AQFKAGwASA%3D#referrer=https%3A%2F%2Fwww.google.com&_tf=From%20%251%24s&share=https%3A%2F%2Fwww.forbes.com%2Fsites%2Fcognitiveworld%2F2020%2F04%2F16%2Fedge-ai-is-the-future-intel-and-udacity-are-teaming-up-to-train-developers%2F

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server component is presented by providing the full-fledged hypervisor [5].

- 2) There were some difficulties in the IDE of the edge computing, and for erasing this problem, the Container based solution is proposed using remote debugging at the edge. This container provides the app developers, the ability to write codes in a productive environment. It also increases the development pace and improves the in-place debugging. In this container, a specific port SSH server is used, which can debug the applications remotely on the edge computing device [6].
- 3) The existing system (Ecosystem Cloud-Centric), could not handle the IoT applications. And its main drawback was that it didn't respond at the real-time. So, to tackle this issue, an IoT device is presented, which processes physical data and manages the devices and gives real-time response. There are a lot of benefits of edge computing; a device was placed in roadside vehicles so that whenever there is a work going on the road, it would be able to lower the noise and focus on a specific voice. And it has integrated edge computing and IoT devices. They made these data validation, metadata annotation [7], and processing of local data based on predefined policies. This device also enables one to measure the temperature of a city. Two cases are mainly discussed in this paper, one of which is mentioned earlier, and secondly a prototype will be introduced for the autonomous cars [8].
- 4) Currently, the smart manufacturing computing framework is facing many problems like the lack of an effective framework of resource scheduling to ensure the low latency requirement. So, a hybrid computing framework is proposed and designed a resource scheduling strategy that's intelligent and thus it can fulfill the real-time requirements in smart manufacturing with support by edge computing. A two-phase algorithm is meant for the system for scheduling the computing resources within the edge layer. And at the end, a prototype was built and experimented upon. With edge computing, it presented the exceptional real-time satisfactory and energy consumption performance of computing services [9].
- 5) Two protocols coded MUDP and MUDP introduced for mobile devices to download the videos, because the mobile data traffic is increasing by 79% traffic. Out of this 79, 60% is just for the video streams and the downloads. The multi-source transmission protocol will transfer the encoded packet from different servers to one client. And MUDP which will encode the RLNC packets. It will be applied to the transport layer, and will also be applied where the server will not cooperate. This protocol will reduce the traffic [10].
- 6) As we know that a huge amount of data is processed in the IoT, and this data puts a lot of burden on the cloud; to reduce this burden, the edge computing is introduced. An edge computing device is proposed; in this device, the DCU (Data Communication Unit) is integrated with some computing devices like the raspberry pi. The ECD is applied in the UWB system, which estimates the tag position and gives response in a short period so that the network's burden may be reduced [11].
- 7) The current progress in IoT is a centralized set-up, where all of the data is sent to the cloud; which had some problems, such as the central point of failure, and it also wastes a lot of bandwidth. To counter this situation, the edge computing was put into use, where most of the data is processed at the edge, and some of the data is sent to the cloud. But there is a need to control the utilized resources at the IoT network, because the computational resources at the edge devices are very limited. Due to the dynamic nature of the IoT, too much uncertainty arises and due to that problem, this design is not suitable for the latency-sensitive applications' [12] requirements and the privacy of the user's data. To overcome these problems, a decentralized resource management framework is proposed. This framework aims to deploy the IoT applications over the edge of the network based on a set of objectives. These sets of objectives include service level agreement (SLA) and the privacy of the data (to achieve satisfiability in the privacy of data). The contribution in this framework is done in three ways:
 - a) The resources would be shared among the nodes based on the application requirements using the task allocation technique.
 - b) All of the participants of the networks would be able to use the available resources according to their requirements.
 - c) If the connection is not stable in the system, there is a dispatcher in the system which works as the handler, or, say, the owner of the system; this dispatcher manages the accessibility of the nodes and other tasks of this like; so, even if the system's connection is not stable, that dispatcher would still operate [13].
- 8) An Edge Fabric Environment named framework is proposed. This framework uses multiple machine learning techniques. Every node is connected and each node is given a task in this system. That network would be plug and play, an example of this system is a wearable device [14] such as a smartwatch. The people/users wearing this wearable device, if they are driving, for example, the watch would automatically connect to their home devices like laptops. This system is connected to the home server using cloud computing. Edge Fabric Environment has the following characteristics:
 - a) It is a decentralized network, upon which, the IoT devices would receive the requests and perform the tasks in response to those requests.
 - b) The framework would join only the trusted IoT devices, and give information about the available resources.
 - c) The IoT devices would keep updating the resources.

- d) IoT devices would be powered by machine learning techniques, which would predict the behavior of other nodes.
- e) The purpose of this device is to lower the latency and minimize the bandwidth [15].

II. METHODS

- 1) HMEC (i.e. the novel concept) is an extended version of the MCC framework. MCC had some flaws in it, and to overcome those flaws, the HMEC is introduced. The model of HMEC still includes a client and a server. But this is just conceptual. In one instance mobile devices act as a server, and in another instance, these act as client.

The use of different methods is illustrated to implement the novel concept, i.e. for the implementation of the HMEC. Several methods are presented graphically to show the comparison between them in terms of the security, application of it in the corporate and private users, and difficulty in adopting, etc. And finally, Intel SGX [16] and the Certificate-based approaches (although they also had some limitations but), were considered the most preferred ones.

The implementation is done using the supernodes which are, actually, the java components. A p2p network is formed by connecting these nodes using TCP and UDP ports. Base of implementation is a distributed hash table (i.e. DHT), the P2P network. The TCP and UDP are also open for the other supernodes, so that those nodes can also connect to this P2P network. To connect mobile devices and computing units a WebSocket port is available. The service that is connected to this port, it manages the computing units for the mobile devices. The service-connected to this port manages the computing units for the mobile devices.

Mobile devices act as the client component, and client framework was developed for this purpose. Client framework emphasizes on the interoperability. Using Dart programming language, this framework is based on web technologies, so all the web environments be able to execute it using the Dart-based cross-platform framework.

The computing unit itself works as the server component. The client framework is again used by that server app, but then it puts the application in a different kind of operation mode. This implementation method does not offer a complete hypervisor for control and nor does it sandbox every offloaded task.

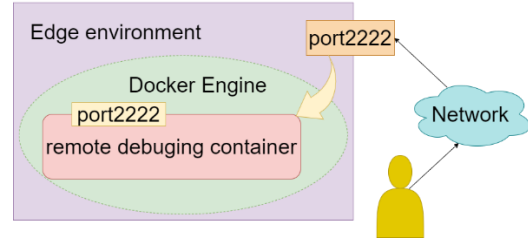


Figure 1. Architecture of the container method.

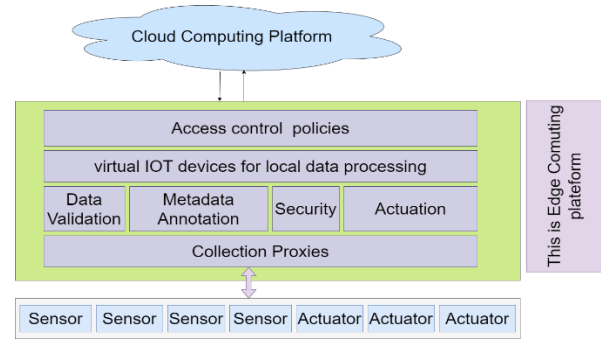


Figure 2. The edge computing architecture.

The global access node (i.e. GAN) is run on the supernode component and fits itself flawlessly in the P2P network. It is operated globally at the user's site. The computing units can be set-up such that to accept merely the offloading requests from previously configured mobile devices. For the proof-of-concept implementation, the certificate-based approach was considered the preferred one and it was enhanced by adding the trusted certificates to it dynamically, using the data security policies [5].

- 2) To counter the stated challenges, a remote debugging method (Fig. 1) is implemented. In this method, a docker container is created, in which all those software packages and libraries are placed, which are used for the development and runtime. Other applications can also be compiled and executed. This remote debugging container can only be used for the runtime. Using this proposed method, IoT applications' performance can be tested in the production environment. And security can be made sure of in the edge devices using an encrypted connection. Because applications are debugged and tested in the real devices, machine data can easily be obtained using the CNC program. And the rest of the dependencies to the edge applications are also tested in the production environment through which less time and a minimum level of effort is consumed.

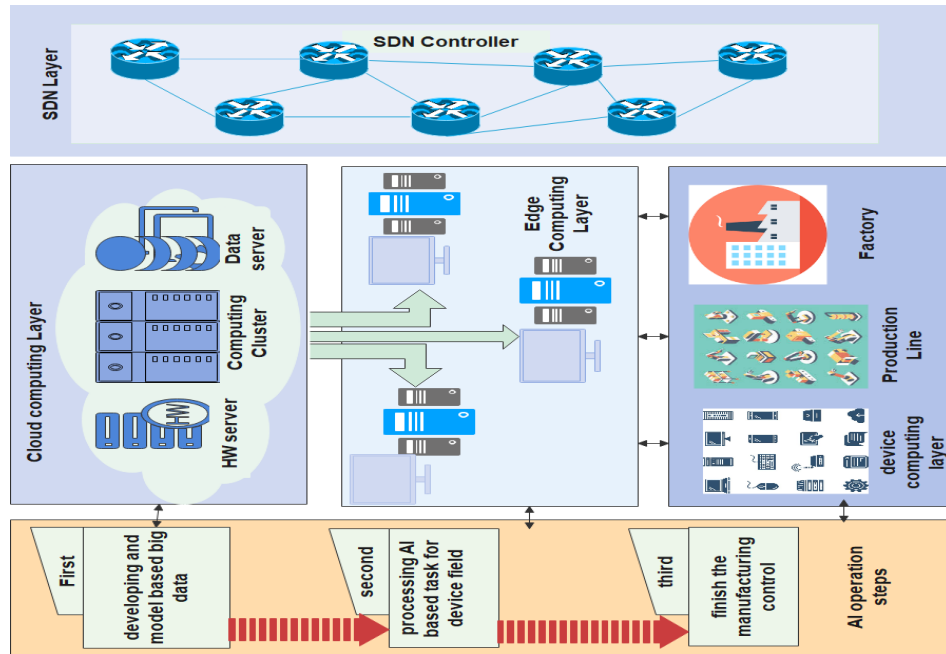


Figure 3. Hybrid computing system architecture.

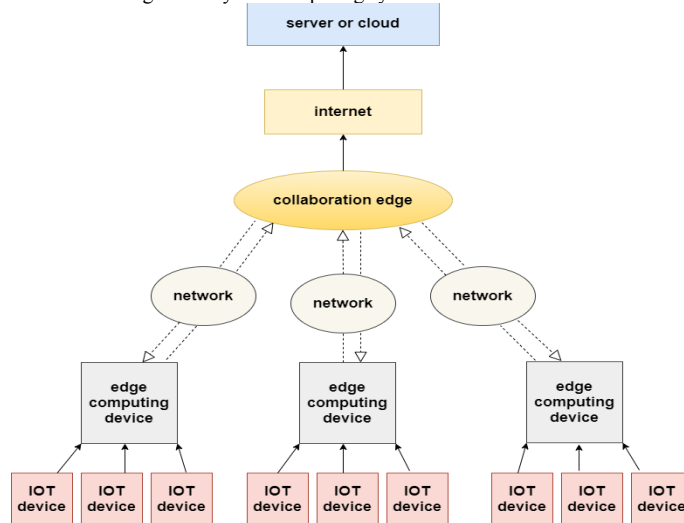


Figure 4. Collaborative edge computing system comprised of several ECD

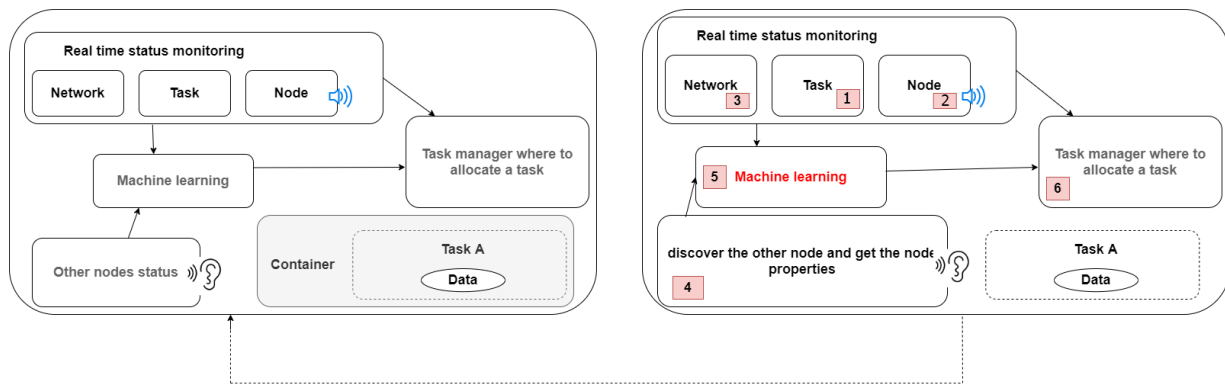


Figure 5. Key components of the DEFT framework

TABLE I. IN THIS TABLE, A PEAK VIEW IS GIVEN, IN WHICH THE DIFFERENT PROBLEMS ALONG WITH THEIR SOLUTIONS PRESENTED AS THE NOVEL IDEAS, AND ALSO THE TECHNOLOGIES USED TO BUILD THOSE IDEAS, AND THE ADVANTAGES OF EACH IDEA ARE DESCRIBED

Comparative Analysis of Edge Computing			
Problems	Solutions	Tools and Technologies	Advantages
Some issues created in edge computing like security (because there is no central device anymore), and performance and battery issues in the mobile devices [5].	A new Hybrid Mobile Edge Computing (HMEC) model is presented [5].	Using the supernodes which are actually the java components. These nodes are connected using the TCP and UDP ports forming a peer-to-peer network [5].	1. The applied strategy is that, the performance cost model would be 70% and the energy-saving cost model would be 30%, as long as the battery is above 50%. And then when the battery is below 50%, the performance cost model starts to decrease gradually to extend the battery lifetime. 2. There is also this improvement in HMEC that, the trust intermediary unit is made between the mobile device and the processing unit utilized for offloading [5].
Problems exist in application development in the domain [6].	The Container based solution using remote debugging at the edge. This container provides app developers, the ability to write codes in a productive environment [6].	It is done within the edge devices, instead of the need to create a new container. To map between this container and the outside world, port 2222 is used. And SSH server is set inside the container [6].	1. use case CNN (Convolutional Neural Network), whose purpose is to check the performance of the computer's vision-task. 2. sequential image layer downloading 3. Decompression 4. I/O Pipelining [6].
The current system (Ecosystem Cloud-Centric), it could not handle the IoT applications. And its main drawback being that it didn't respond at the real-time [8].	An IoT device, which processes physical data and manages the devices and gives real-time response [8].	3G, Wi-Fi, and BLE are used for the communication, and the protocols are HTTP and CoAP (Constrained Application Protocol). HTTP works as TCP protocol and CoAP works as UDP [8].	This device would provide the roadside assistant system; by using that system, the consumers would be able to speak and their voice would be sent to the service providers, and that way the consumers will get the quick response [8].
Traffic issues increase in the browser [10].	The two protocols coded MUTP, and Multi-source transmission protocol are introduced for the downloading of videos on mobile devices [10].	The Coded MUTP is made through the network coding to transfer the encoded multiple data. An Origo server is used for downloading [10]	The Uncoded MUTP works double fast than the Parallel activity. And the coded MUTP works three times faster than the HTTP and 25% faster than the Uncoded MUTP for downloading the video; because these are using the Origo server [10].
A huge amount of data is processed in the IoT, and this data puts a lot of burden on the cloud [11].	Edge computing device is proposed [11].	The ECD devices are connected directly to the server. Five sensors are attached in the ECD, namely, UWB, RFID, scale, temperature, humidity [11].	ECD device, the burden on the cloud is reduced, and also, it filters the data and then sends it forward [11].
The edge devices are very limited. Due to the dynamic nature of the IoT, too much uncertainty arises and due to that problem, this design is not suitable for the latency-sensitive applications' requirements and the privacy of user's data [13].	A decentralized resource management framework is proposed [13].	Z3 SMT solver tool is used [13].	This framework utilizes the available resources easily and efficiently; add to that, it also guarantees the privacy protection [13].
The previous system was based on a centralized approach, due to that, only the closest nodes would get the resources' updates; and the nodes that were at a distance, they'd get the late response. The new system is based on a decentralized approach. The old system would only support mobile devices, whereas the current system is made compatible with all types	Edge Fabric Environment named framework is proposed. This system is connected to the home server using cloud computing [15].	UDP, TCP, MQTT protocols are used. Machine learning is used as language and the tools that are used for ML are lasso regression, and ridge regression [15].	Using the help of a CPU test, the time of nodes would be decreased from nearly 35000 seconds to under 8000 seconds. The system would select those nodes which would be the best candidate based on prediction [15].

Comparative Analysis of Edge Computing			
Problems	Solutions	Tools and Technologies	Advantages
of devices and the platforms. The old system had limited battery issues, which are taken care of in the new system [15].			

And any new change that is made through this method, it is done within the edge devices, instead of the need to create a new container. To map between this container and the outside world, port 2222 is used. And SSH server is set inside the container (Fig. 1). This server provides a secure connection to protect the file-sharing process [6].

- 3) 3G, Wi-Fi, and BLE are used for the communication, and the protocols are HTTP and CoAP (Constrained Application Protocol). HTTP works as TCP protocol and CoAP works as UDP. CoAP is used to reduce the latency and it consumes less power, and HTTP protocol is used to transmit the data virtually on the World Wide Web. Fig. 2 portrays the edge computing architecture [8].
- 4) A prototype platform (Fig. 3) was built to evaluate the proposed strategies of scheduling the edge computing resources. This platform was made up of the four main components, these are device field, edge computing layer, industrial private cloud servers, and industrial networks (with Software-defined networking).

The device field contained different kinds of equipment like mobile robots, product processing machines, conveyor belts, and manipulators, etc. To implant the edge computing layer, single-board computers (SBC) were connected with the LANs and the Raspberry Pi 3 Model B to work as the edge servers. To create a link between the servers and the devices, the ethernet and Wi-Fi connections were installed. The Linux OSs and OpenCV frameworks were installed in the edge servers for the job of image processing in the mobile robots. A private cloud platform was built to back the big data processing using the XenServer and Hadoop cloud ecosystem. So that made up the whole prototype platform ready to engage in.

A mobile robot was used to capture the industrial operations of the equipment. An industrial camera was fit with that robot to capture the footages and then it would calculate the boundary of the image for the next work process. After that, a neural network was used to measure the overall state of the equipment. At the end, the images were sent to the computing layers, via industrial networks, for examining the performance of the practiced methods.

In the meantime, the computing latency, satisfaction degree and the energy consumption, these three are used as the assessment metrics for assessing the provided strategies by selecting algorithms for ES (SAE) and cooperating edge computing servers. The proposed strategies' performance is examined in comparison with the traditional strategies [9].

- 5) MUDP protocol is made, which is used to download the videos. In this, the Coded MUDP is made through the network coding to transfer the encoded multiple data. An

Origo server is used for downloading. To manage the proxy, the video will be downloaded in either of these 3 different ways, i.e. through parallel HTTP, or Uncoded MUDP, or by coded MUDP protocols. Parallel HTTP will work only through different TCP. The channel that the Uncoded MUDP and Coded MUDP will create, these will download fast by connecting the unreliable and different proxies.

The main job of Uncoded MUDP is that it maintains the client list and handles the packets through DoF. And the Coded MUDP doesn't take the receiver and the transit data in the form of packets, rather, in the form of generation [10].

- 6) The ECD devices that are introduced for IoT, the raspberry pi is used in this device to reduce the time and cost. It would calculate the data and give a quick response to it in real-time. The DCU is used for the interface of sensors. The UWB in the system, is used to trace the cars or any IoT based devices. All of the data would be sent directly to the DB MySQL system, and it displays the data in a fast manner. The ECD devices are connected directly to the server, which lowers the burden. Five sensors are attached in the ECD, namely, UWB, RFID, scale, temperature, humidity; these are all used for the process and sending of the data. Fig. 4 depicts the proposed system architecture [11].
- 7) The SMT formula is used to complete the required task. It has the following five parts:
 - a) Task facts: These are a set of rules which would place the task correctly on the edge.
 - b) The second and third task is domain facts and latency facts. These both are combined into one, and are used to map the encoding communication constraints
 - c) Bid constraints: These constraints are used for the dispatcher to select the valid solution from available resources.
 - d) Formulation Constraints: These rules are used to validate the solution that was selected by the dispatcher [13].
- 8) UDP, TCP, MQTT protocols are used for the communication. UDP is used for the discovery of new nodes, and also for updating the status of the nodes. Machine learning is used as language and the tools that are used for ML [17] are lasso regression, and ridge regression. Fig. 5 explains the DEFT framework. There are different nodes and every node has the job as a task requester or a task receiver. In Fig. 5, two nodes are presented, the right one is task requester, and the node on

left side is task receiver. The task requester node (i.e. the node on right side) will first identify the type and objectives of that task. As it is shown in Fig. 5 that there is task numbered with 1 in the right hand side node, and at the same time, the task requester node will check its available resources such as CPU, memory, battery, etc. and it will also check the status of the network. After doing all that, the requester node will look for the nearest node that would be able to do its task. All of this process would be done using the machine learning algorithm. The requester node would send all of the gathered data to the nearest node; then the receiver node would run an analysis on that data and send back the required information to requester node [15].

III. RESULTS

The introduction of a flexible structured network is the noticeable upgrade. In this system, the computing units and mobile devices can connect with one another by signing in, enabling it to dispose of the tasks by bringing closer the mobile devices; and processing units, offering their resources for computation.

Since this system is designed for the mobile devices, this edge computing environment is very useful for the user in that context. Because the user tends to be on the move mostly, sometimes on-road or maybe on the fly, wherever the user is, he will need the computing units to do the data processing, meaning he will need the internet all the time. So, in the HMEC, some of the computing units and storage are kept in the background tasks closer in proximity to the device.

Battery is improved in this system. The applied strategy is that, the performance cost model would be 70% and the energy-saving cost model would be 30% as long as the battery is above 50%. And when the battery is below 50%, the performance cost model starts to decrease gradually to extend the battery lifetime.

There is also this improvement in HMEC that, the trust intermediary module/component is built in between the mobile device and computing unit used for the offloading. This component establishes trust between the two [5].

First, the effort was made to measure the suitability of EC's IoT application, in which, the issue was that the network bandwidth was limited. In another paper's challenges, the discussion was done on the EC's opportunities.

A use case CNN is presented (Convolutional Neural Network), whose purpose is to check the performance of the computer's vision-task. In the container domain, 3 different optimizations are combined by the Docker-Pi, to reduce the deployment time. There are 4 factors in it, 1: sequential image layer downloading, 2: Multi-Threaded Layer, 3: decompression, 4: I/O Pipelining. But even then, remote debugging and deployment for ED were not well addressed. So, this research work is completed using these papers [6].

The edge computing system will support the IoT devices through the Raspberry pi3, through which the IoT devices will exchange the data. The platform would be like the scenario of a smart city. The devices would be attached to the vehicles, through which the temperature of the city would be measured. This device would provide the roadside assistant system; by

using that system, the consumers would be able to speak and their voice would be sent to the service providers, and that way the consumers will get the quick response [8].

The analysis and results show that, the increment in the task data size, causes for all the methods to increase the computing latency. By any means, the proposed system will outperform the other techniques. Add to that, if the data size increases, our algorithm (SAE_CEC) will exploit multiple edge servers simultaneously to easily process the tasks requiring extensive-computing. If compared with CSC, the OEC has better performance because it adopts edge computing servers for finishing the tasks. Whereas, in the case of CSC, it has worst performance when it comes to computing latency; since the computation servers are far away from the device, it causes the system to spend extensive time in data transmission.

SAE_CEC also performs better because it can select the better edge servers dynamically following the deadline of the task.

Conclusively, this proposal can adapt itself to the different real-time constraints and varying environments. The legacy algorithms cannot be implemented in the modern-day industrial systems due to the failure of fulfilling the real-time requirements [9].

The result of these three works the same way, but the Uncoded MUTP works double fast than the Parallel activity. And the coded MUTP works three times faster than the HTTP and 25% faster than the Uncoded MUTP for downloading the video, because these are using the Origo server [10].

With the ECD device, the burden on the cloud is reduced, and also, it filters the data and then sends it forward. It doesn't send the unwanted, unnecessary, and raw data over the network. It uploads the result in a real-time fast pace [11].

To measure the performance of the framework, the dispatcher was deployed on the single-core Intel i5 2.5 GHz processor. Edge architectures of different sizes were performed a hundred times, which presented different results, and based on those results, it was revealed that this framework utilizes the available resources easily and efficiently; add to that, it also guarantees the privacy protection [13].

Using the different protocols and tools that are mentioned above in the methods section, a system is proposed, through which, the nodes would be chosen based on predicted time. Using the help of a CPU test, the time of nodes would be decreased from nearly 35000 seconds to under 8000 seconds. The system would select those nodes which would be the best candidate based on prediction. The network would be plug and play, through which new nodes would enter the fabrics [15].

IV. CONCLUSION & FUTURE WORK

The edge computing is like a new norm in the IIOT's (Industrial Internet of Things') [18] application developments. But there are many challenges in its implementation. Those challenges were found and a solution is presented. For this purpose, the remote debugging container method is presented. And by applying this method, significant results were revealed regarding the development time and security. Aside from that, any application can be used remotely and securely in the edge devices by running port2222 and SSH server in the container.

This study is suitable for the remote development of C++ applications [6].

Edge architecture is presented and the aim is to manage the IoT devices and process the local data. The virtual IoT devices would give a response through actors in real-time. The architecture would work on two things, on the roadside and in counting vehicles [8].

The proposed protocols (i.e. Coded MUTP and Uncoded MUTP) would be used to increase the performance. These protocols work double fast than the HTTP protocols, and take less bandwidth for the download. Also, it helps stream the video fast and with less bandwidth [10].

The proposed edge computing device traces the IoT devices and also filters the data, and UWB gives responses in real-time. This system helps in sharing the data to IoT based system and also helps to ease off [11].

A decentralized resource management technique and technical framework is proposed for the deployment of latency-sensitive edge network's IoT applications. This framework can utilize the available resources efficiently and also ensures the privacy protection. And this framework can also find any other solution, if there might be [13].

In the dynamic edge fabric environment, multiple machine learning techniques are used for task allocation. This framework would be implemented and tested using heterogeneous IoT devices. The allocation of resources would be dynamic and it would use machine learning. Regression methods would ensure the accurate task deployment. Prediction models are very beneficial, through which the cross-validation would be done [15].

As this method is suitable for the remote development of C++ applications, similarly, it would also give support for the Python, Java, and NodeJS [6]. In the future, work can be done on doing more research on these protocols' (Coded MUTP and Uncoded MUTP) packet scheduling system, and it will make the system more efficient [10]. In the future, execution time would be reduced and the encoding would be improved, and also the number of participant nodes in a network would be increased [13].

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